Home teams win more. Do the fans explain it?

Conventional wisdom: KBO clubs play better at home, and a packed stadium is what lifts them. **H_0:** $\mu(\Delta WR) = 0$, where ΔWR = home win rate − away win rate for the same team-season. **H_1:** $\mu(\Delta WR) > 0$ *and* the edge grows with attendance ($\beta_{attendance} > 0$). **Data:** 152 team-seasons (2010–2025, home/away records and batting/pitching splits) plus 3,494 games (2021–2025) with crowd, weather and calendar data.

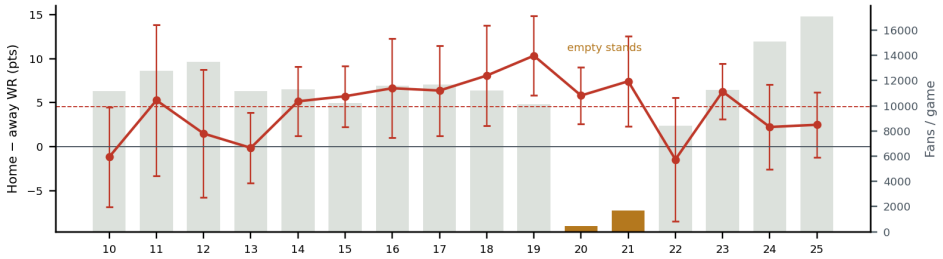
HOME EDGE
REAL · +4.5 pts
95% CI +3.4 to +5.7, $p < 0.001$; 73% of team-seasons better at home.

CROWD EFFECT
NOT FOUND
−0.13 pts per 1,000 fans (CI −0.42 to +0.17), $p = 0.40$. Not proof of zero.

TEAM DIFFERENCES
LOOKS LIKE LUCK
Spread SD 7.4 pts vs 8.4 from coin flips alone (χ^2 $p = 0.99$).

KEY FINDINGS

QUESTION	METHOD	ESTIMATE	95% CI	P	VERDICT
Is there a home edge?	Paired t (n = 152)	+4.5 pts	+3.4 to +5.7	<0.001	**Reject H_0**
Better hitting / pitching at home?	One-sample t	OPS +0.022 · ERA −0.20	both exclude 0	<0.001	**Yes**
+1,000 fans → bigger edge?	Simple OLS (n = 132)	−0.13 pts	−0.42 to +0.17	0.403	**Not significant**
Same, 15 confounders removed	Double ML	+0.47 pts	−0.10 to +1.03	0.105	**Not significant**
Edge with near-empty stands (2020)?	One-sample t (n = 10)	+5.8 pts	+2.6 to +9.0	0.001	**Intact**
Shared stadium (LG, Doosan) vs others	Welch t	−3.5 pts	−6.3 to −0.7	0.016	**Nominal only (Holm p = 0.20)**
Jamsil derby (both teams at home)	Two-proportion z, 77 games	48.1% vs 53.8%	−17.6 to +6.1	0.343	**Right way, too few games**
Weather, weekday, holiday, start time, crowd (12 features)	χ^2 across groups, 3,494 games	0 of 12 significant	–	all >0.17	**None significant**
Which X predicts ΔWR best?	Simple OLS, R^2	OPS gap 26.4% vs attendance 0.5%	–	<0.001	**Performance, not crowd**



League-average ΔWR by season with 95% CI (red) and average home attendance (bars; orange = COVID seasons). The edge held in 2020, when stands were closed until late July and then only partly open.

WHAT THE DATA SHOWS VS. CONVENTIONAL WISDOM

The first half of the belief is right: home teams win about 4.5 points more often, in every era, because they hit better and allow fewer runs at home (together explaining 32% of ΔWR). For the second half we found no evidence in any test: simple, controlled, within-team, occupancy split, Double ML with 15 confounders, and 3,494 individual games. “Not significant” is not proof of zero: the attendance interval still allows a small effect (up to +0.17 pts per 1,000 fans). The natural experiments point toward the **place** more than the people: the edge survived near-empty stands in 2020, is smaller for the two clubs that share Jamsil ($p = 0.016$, but not significant after Holm correction for our 22 tests), and falls to 48% when those two clubs play each other, although that last test (77 games) is not yet significant.

IMPLICATIONS

(1) Front offices: justify attendance marketing on revenue and fan experience, not on wins. **(2) Forecasters:** treat home field as a near-constant 4–5 point bonus; team-to-team differences are consistent with luck, and crowd size added no predictive value in our data. **(3) League and city:** LG and Doosan carry a structural home handicap while sharing Jamsil; any stadium redevelopment should be evaluated for its effect on both clubs' home edge.

Limits. 22 tests run: after Holm correction only the home edge, the home/away splits and the 2020 result remain significant. Team-season aggregates for most tests; team-level features vary across only 10 clubs; capacities are nominal; 77 derby games are too few for significance. **Next:** more derby seasons, umpire ball/strike calls with and without crowds, per-game rest and travel.

Provenance: W-L records cross-checked 2014–2025 against MyKBO Stats and Wikipedia (119/119 match); game rows carry IDs, dates, teams and scores; weather is measured KMA ASOS data at the first-pitch hour; occupancy is estimated from nominal capacity. Interactive webpage, raw data (team-season and game-level CSV) and code repository are linked in our Slack and LXP submission. Sources: KBO records via Yagoonara and MyKBO Stats; KBO official attendance; kbo-crowd game logs with KMA weather; Wikipedia/KMA stadium, census and climate data; Jeong (2023), SKKU thesis.